

Tanmoy Debnath

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EDUCATION

Techno College of Engineering Agartala

2021 – 2025

Bachelor of Technology in Computer Science and Engineering

EXPERIENCE

Junior Software Engineer

Jun 2026 – Present

xLayer Technologies

Agartala, India (On-site)

- Collaborate with cross-functional teams in agile sprints to develop, maintain, and scale full-stack web applications.
- Develop and optimize REST APIs using Tornado, improving system reliability and reducing average endpoint response times.
- Design responsive user interfaces in Angular and document technical specifications to streamline API integrations.
- Manage schema designs and optimize PostgreSQL and MongoDB database queries to accelerate data retrieval speeds.

Automation Engineer & Backend Developer

Sep 2025 – May 2026

Yupcha

Onsite

- Spearheaded automated CI/CD pipelines using GitHub Actions and Docker, accelerating deployment velocity by 40% and ensuring zero-downtime releases.
- Pioneered computer vision pipelines leveraging PyTorch and YOLOv8 to process 50k+ images daily with 95% classification accuracy for autonomous systems.
- Streamlined RESTful API endpoints using FastAPI and PostgreSQL indexing, sustaining 50k+ daily requests with sub-50ms latency metrics.
- Maximized system reliability by deploying high-availability protocols and real-time monitoring, increasing uptime to 99.9% and reducing resource consumption by 25%.
- Authored event-driven Python automation modules to synchronize real-time data across distributed services, boosting system throughput by 35%.

PROJECTS

Blockchain Certificate Verifier (Cyphire)

2026

FastAPI, Solidity, Polygon, PostgreSQL, Web3.py | GitHub

- Platform for issuing immutable, digitally-signed certificates anchored on the Polygon blockchain.
- Forged custom Solidity smart contracts on Polygon network, facilitating immutable record-keeping and instant verification for 1,000+ certificates.
- Engineered asynchronous Web3.py middleware within a FastAPI architecture, reducing blockchain transaction latency by over 90%.
- Codified a dual-verification protocol synchronizing PostgreSQL databases with Polygon blockchain consensus to guarantee 100% data integrity.

Link Ingestor API

2025

FastAPI, Celery, Redis, BeautifulSoup | GitHub

- High-performance autonomous system for large-scale webpage data ingestion with async processing architecture.
- Engineered autonomous web crawling modules using Bing Search API and BeautifulSoup, achieving full automation of data extraction workflows.
- Deployed distributed task queues using Celery and Redis to balance 200+ concurrent ingestion streams, eliminating architectural bottlenecks.
- Parallelized data ingestion flows via multi-threaded execution, reducing average page processing time by 60% for high-traffic environments.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, C/C++, Solidity, HTML/CSS

Backend & Frameworks: Angular, Tornado, FastAPI, Node.js, WebSockets, REST APIs, Celery

Machine Learning: PyTorch, YOLOv8, ResNet, Computer Vision

Databases & Storage: PostgreSQL, Redis, MongoDB, SQLAlchemy, SQLite

DevOps & Cloud: CI/CD (GitHub Actions), Docker, Infrastructure as Code, Linux, Git

Blockchain: Web3.py, Smart Contracts, Polygon (EVM), Decentralized Verification

Tools: RabbitMQ, Postman, VSCode, Bash/Linux Terminal, Distributed Systems, Scalability